

 [Learn](#)  [Practice](#)

Chemical bonding & hybridisation — Issue #33 candidate: Construction

Contents and concept route

Concept Book

Hybridisation, sigma/pi overlap and molecular shape

Learning promise

Introductory localized hybrid-orbital descriptions for the supplied BF_3/NH_3 , H_2CO , ethene, CO_2 , CH_3F , HCN , methylamine and $\text{CH}_4/\text{NH}_3/\text{H}_2\text{O}$ cases.

Concept route

1. Domain count before hybridisation label
 1. Test whether three sigma bonds are sufficient for the proposed label.
 2. Separate number of bonding directions from bond identity or strength.
 3. Choose the local-domain ledger before the hybrid label
2. Sigma/pi orbital inventory without double-counting
 1. Conserve carbon orbitals while separating domains from sigma/pi components.

2. Construct the relation between the molecular plane and the pi orbitals.
 3. The central carbon inventory must support two directions and two pi components.
 4. Keep local inventories separate from the triple-bond component inventory.
 5. Avoid spending a carbon basis function twice.
3. Electron-domain geometry versus molecular shape
1. Compare full local inventories, not only the common three sigma bonds.
 2. Count local domains and bonded nuclei at two different centres.
 3. Test whether domain geometry and nuclear shape are interchangeable.

Outside this book

- Delocalised molecular-orbital treatments, hypervalent models, radicals and amide resonance beyond the supplied questions.

Sections

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Test whether three sigma bonds are sufficient for the proposed label.

Separate number of bonding directions from bond identity or strength.

Choose the local-domain ledger before the hybrid label

Domain count before hybridisation label

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Domain count before hybridisation label

Concept difficulty: Medium concept

1

Test whether three sigma bonds are sufficient for the proposed label.

Medium concept

Next →

Section 1 of 11

Conditions for this construction

- Can draw simple Lewis structures.
- Can count central electron domains.
- Knows introductory 2/3/4-domain geometry rules.

Why this section matters for the question set

—

This unit builds toward

- **Q5** · D2 · the hardest question in this set
The move learners miss: Use nitrogen as the counterexample.
[Try it in Core2](#)

1 Hold the claimed three-sigma condition fixed.

Why valid: Both supplied centres satisfy it, so an omitted lone-pair condition can test the implication.

Result: BF₃ and NH₃ both have three sigma bonds.

2 The step the hard question turns on Use nitrogen as the counterexample.

Why valid: NH₃ also has one central lone pair; its local total is four.

Result: Three sigma bonds alone do not imply three domains/sp².

3 Replace the claim with a conditional local rule.

Why valid: Within localized introductory classification, bonds plus central lone pairs determine the domain total.

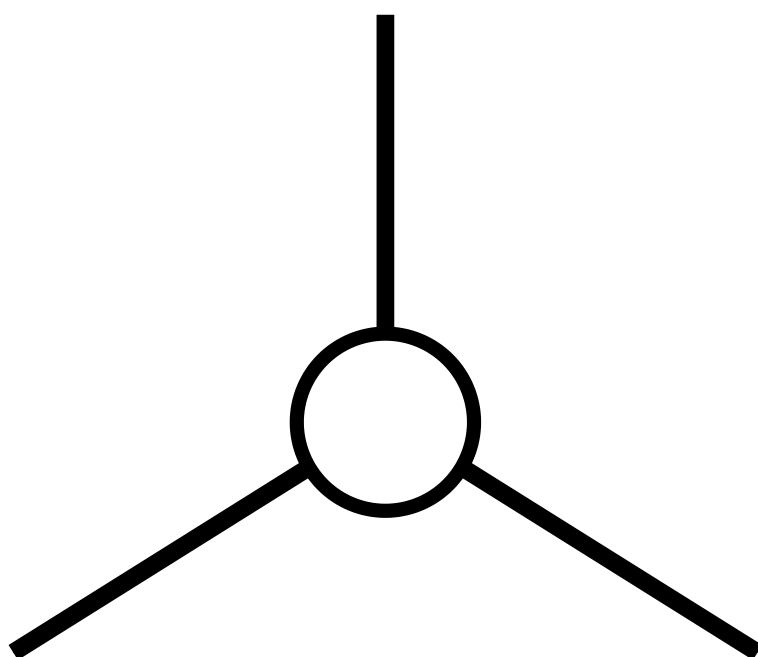
Result: Use the full inventory: three domains -> sp² model; four -> sp³-like model.

4 Separate shape observations from a unique orbital basis.

Why valid: Nuclear geometry constrains a description but does not measure hybrid coefficients.

Result: The counterexample rejects the bond-only rule without proving unique physical hybrid orbitals.

Count directions / regions first



3 domain directions

Count regions before labels.

Stage 1 of 3

Separate electron-domain directions from atom-only molecular shape and make lone-pair occupancy visible.

The key step

BF₃ has three B domains; NH₃ has three bonds plus one N lone pair. Introductory sp²/planar versus sp³-like domain/pyramidal shape follows the local model.

Worked explanation

Would a three-bond nitrogen with a lone pair and a three-bond boron without one require identical domain inventories? Explain.

Follow the operation, its justification and its result.

1. A counterexample must hold the claimed condition while changing an omitted relevant condition.
2. No: four versus three. Keeping the answer identical despite the changed lone-pair condition suggests the misconception.
3. NH₃ falsifies three-sigma-implies-sp²: its local lone pair makes four domains. Geometry constrains a model but does not uniquely prove a hybrid basis.

Result

No: four versus three. Keeping the answer identical despite the changed lone-pair condition suggests the misconception.

Check

Changed check: BF₃ and NH₃ share three bonds but differ in local lone pairs; the proposed rule predicts the same result incorrectly.

Mistake clinic · diagnose and repair

A tempting wrong path

- Three sigma bonds suffice to prove sp².
- An electronegative substituent adds a domain beyond its bond direction.

Diagnose

- Would a three-bond nitrogen with a lone pair and a three-bond boron without one require identical domain inventories? Explain.

- Count carbon domains in CH₄, CH₃F and CH₂F₂. Does each F add an extra domain beyond its C-F direction?

Repair

- NH₃ falsifies three-sigma-implies-sp²: its local lone pair makes four domains. Geometry constrains a model but does not uniquely prove a hybrid basis.
- CH₃F carbon has four bonding directions and introductory sp³ accounting. The count alone establishes neither bond lengths nor their detailed causes.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: BF₃ and NH₃ share three bonds but differ in local lone pairs; the proposed rule predicts the same result incorrectly.

2 • APPLY

Repair the rule with an explicit applicability condition.

3 • CONNECT

A counterexample must hold the claimed condition while changing an omitted relevant condition.

2

Separate number of bonding directions from bond identity or strength.

Medium concept

Section 2 of 11

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q6 • D1

The move learners miss: Apply the same local four-domain allocation.

[Try it in Core2](#)

1 Recount carbon neighbours after H-to-F substitution.

Why valid: CH_3F has three H and one F directly attached to C.

Result: Four carbon bonding directions.

2 The step the hard question turns on Apply the same local four-domain allocation.

Why valid: Changing substituent identity does not itself create a new neighbour or lone pair at C.

Result: Introductory sp^3 carbon model.

3 State what the count leaves open.

Why valid: Bond lengths, polarity and exact angles require information beyond neighbour counting.

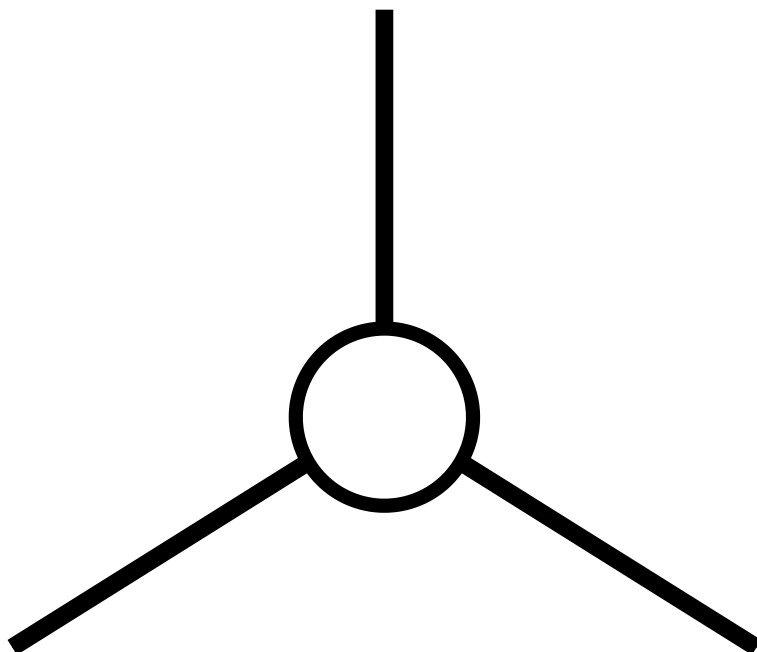
Result: No equality of C-H and C-F properties is inferred.

4 Test CH_2F_2 as a changed inventory.

Why valid: Two H and two F are still four neighbours.

Result: Four domains remain; electronegativity is not an extra domain.

Count directions / regions first



3 domain directions

Count regions before labels.

Stage 1 of 3

Separate electron-domain directions from atom-only molecular shape and make lone-pair occupancy visible.

Worked explanation

Count carbon domains in CH_4 , CH_3F and CH_2F_2 . Does each F add an extra domain beyond its C-F direction?

Follow the operation, its justification and its result.

1. Replacing a substituent changes chemical environment without automatically changing connectivity count.

2. All four; no.
3. CH_3F carbon has four bonding directions and introductory sp^3 accounting. The count alone establishes neither bond lengths nor their detailed causes.

Result

All four; no.

Check

Changed check: CH_2F_2 still has four central bonding directions; electronegativity is not an extra domain.

Mistake clinic • diagnose and repair

A tempting wrong path

- Three sigma bonds suffice to prove sp^2 .
- An electronegative substituent adds a domain beyond its bond direction.

Diagnose

- Would a three-bond nitrogen with a lone pair and a three-bond boron without one require identical domain inventories? Explain.
- Count carbon domains in CH_4 , CH_3F and CH_2F_2 . Does each F add an extra domain beyond its C-F direction?

Repair

- NH_3 falsifies three-sigma-implies- sp^2 : its local lone pair makes four domains. Geometry constrains a model but does not uniquely prove a hybrid basis.
- CH_3F carbon has four bonding directions and introductory sp^3 accounting. The count alone establishes neither bond lengths nor their detailed causes.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: CH₂F₂ still has four central bonding directions; electronegativity is not an extra domain.

2 • APPLY

Classify carbon in CHCl₃ without assigning an exact bond angle.

3 • CONNECT

Replacing a substituent changes chemical environment without automatically changing connectivity count.

3

Choose the local-domain ledger before the hybrid label

Medium concept

Section 3 of 11

← Previous

Next →

1 Inventory bonds and central lone pairs separately.

Why valid: The givens supply three bonds at each centre but different lone-pair counts.

Result: BF₃: 3+0=3 domains; NH₃: 3+1=4.

2 Apply the bounded introductory domain models.

Why valid: The local total, rather than bonds alone, selects the model arrangement.

Result: BF₃: planar/sp²; NH₃: tetrahedral domains/sp³-like.

3 Name shape using bonded nuclei only.

Why valid: The N lone-pair domain has no bonded nucleus.

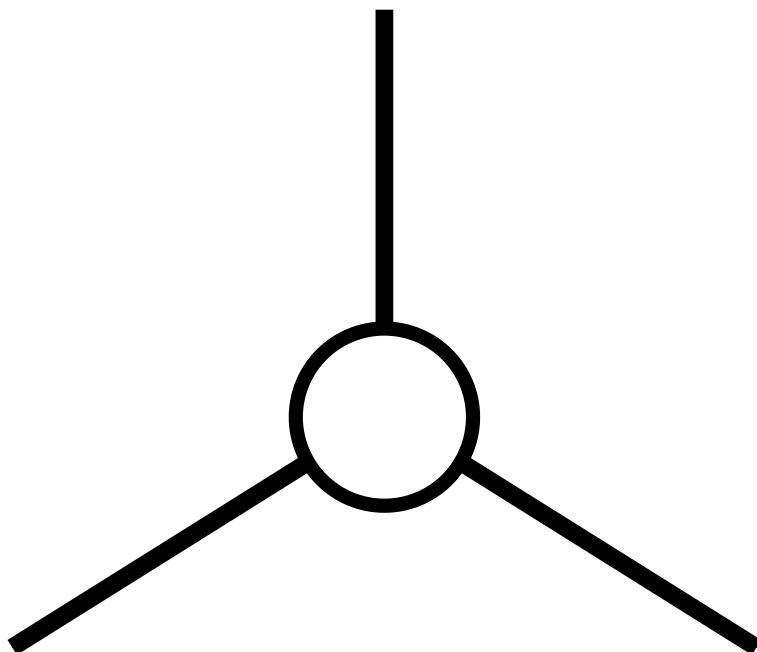
Result: BF₃ is trigonal planar; NH₃ is trigonal pyramidal.

4 Compare NH₃ with H₂O.

Why valid: Three-bond/one-pair and two-bond/two-pair inventories both total four.

Result: The same domain total can give different bonded-nucleus shapes.

Count directions / regions first



3 domain directions

Count regions before labels.

Stage 1 of 3

Separate electron-domain directions from atom-only molecular shape and make lone-pair occupancy visible.

Worked explanation

Count local domains and bonded nuclei separately in BF_3 , NH_3 and H_2O .

Follow the operation, its justification and its result.

1. A central lone pair adds a domain even though it adds no bonded nucleus.
2. BF_3 has three B domains; NH_3 has three bonds plus one N lone pair.
Introductory sp^2 /planar versus sp^3 -like domain/pyramidal shape follows the

local model.

Result

Totals 3,4,4; bonded nuclei 3,3,2. A repeated bonds-only rule differs from an isolated missing lone pair.

Check

Changed check: H₂O and NH₃ have different numbers of bonds but the same four-domain total.

Mistake clinic • diagnose and repair

A tempting wrong path

- Three sigma bonds suffice to prove sp².
- An electronegative substituent adds a domain beyond its bond direction.

Diagnose

- Would a three-bond nitrogen with a lone pair and a three-bond boron without one require identical domain inventories? Explain.
- Count carbon domains in CH₄, CH₃F and CH₂F₂. Does each F add an extra domain beyond its C-F direction?

Repair

- NH₃ falsifies three-sigma-implies-sp²: its local lone pair makes four domains. Geometry constrains a model but does not uniquely prove a hybrid basis.
- CH₃F carbon has four bonding directions and introductory sp³ accounting. The count alone establishes neither bond lengths nor their detailed causes.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: BF₃ and NH₃ share three bonds but differ in local lone pairs; the proposed rule predicts the same result incorrectly.

2 • APPLY

Repair the rule with an explicit applicability condition.

3 • CONNECT

NH₃ falsifies three-sigma-implies-sp²: its local lone pair makes four domains.

Geometry constrains a model but does not uniquely prove a hybrid basis.

Question-specific diagnosis and repair

Repair for Q5

Test the rule on a changed case

Would a three-bond nitrogen with a lone pair and a three-bond boron without one require identical domain inventories? Explain.

Your prediction and reason

Repair for Q6

Test the rule on a changed case

Count carbon domains in CH₄, CH₃F and CH₂F₂. Does each F add an extra domain beyond its C-F direction?

Your prediction and reason

Now you do one

Count C-centred domains in CH₂Cl₂ and separately count all attached atoms. State which bond properties this inventory cannot determine.

Model answer

Four C-neighbour domains; no C lone pair. It determines neither equality of bond lengths nor exact angles.

1. Four C-neighbour domains; no C lone pair. It determines neither equality of bond lengths nor exact angles.

Practice this concept in Core2

Q5

Q6

Sections

Conserve carbon orbitals while separating domains from sigma/pi components.

Construct the relation between the molecular plane and the pi orbitals.

The central carbon inventory must support two directions and two pi components.

Keep local inventories separate from the triple-bond component inventory.

Avoid spending a carbon basis function twice.

Sigma/pi orbital inventory without double-counting

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Sigma/pi orbital inventory without double-counting

Concept difficulty: Hard concept

1

Conserve carbon orbitals while separating domains from sigma/pi components.

Hard concept

Section 4 of 11

[← Previous](#)[Next →](#)

Conditions for this construction

- Can count electron domains.
- Knows introductory 2/3/4-domain geometry rules.

Why this section matters for the question set

This unit builds toward

- Q2 · D2

The move learners miss: Allocate the four carbon valence functions.

[Try it in Core2](#)

1 Count C neighbours in HCHO.

Why valid: Two H and one O are three connections; bond multiplicity is a separate inventory.

Result: Three domains; no C lone pair.

2 The step the hard question turns on Allocate the four carbon valence functions.

Why valid: One 2s plus two 2p form three sp^2 hybrids, leaving the third 2p unused in that mixing.

Result: Three sigma-direction hybrids plus one residual p.

3 Resolve the C=O bond components.

Why valid: A localized double-bond account has one axial sigma and one lateral pi component.

Result: C=O: one sigma + one pi; residual C p supplies the pi role.

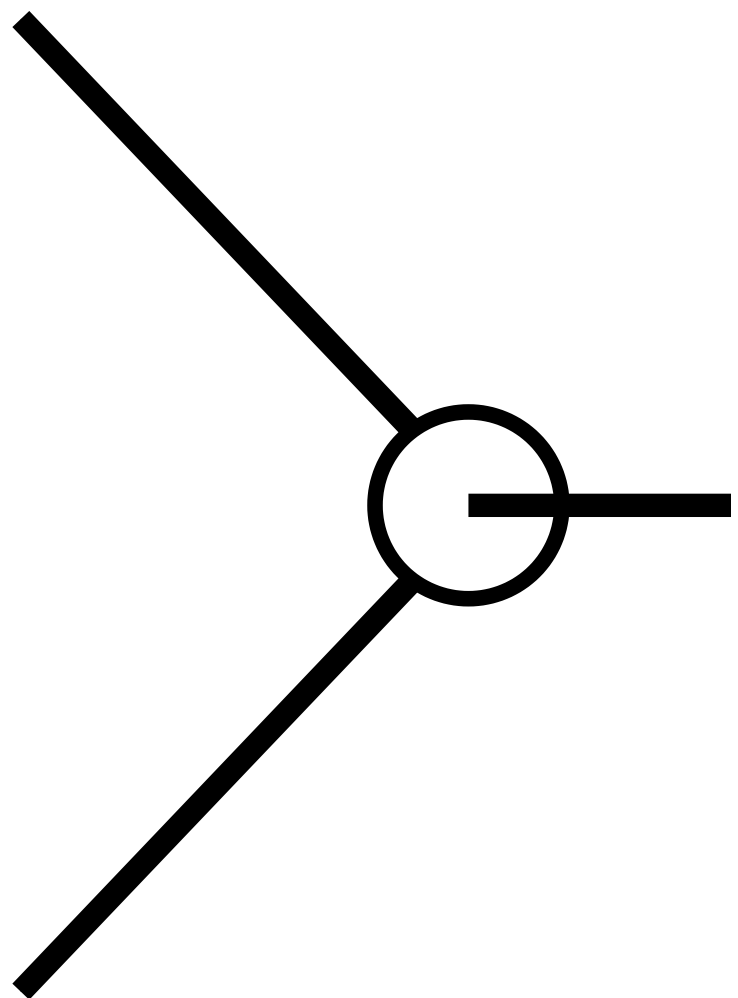
4 Close the basis ledger using an sp comparison.

Why valid: sp gives two hybrids and two residual p functions from the same

four inputs.

Result: Both allocations total four; no p function is spent twice.

Stage 1 — sigma-direction frame



page = molecular pla

Place three directions in one plane.

Stage 1 of 4

Build a generic three-direction sigma frame and add a perpendicular p direction for side-on overlap without reusing orbitals.

The key step

sp² accounting is three hybrid orbitals plus one residual p. A p function already mixed into a hybrid is not simultaneously counted as an additional residual orbital.

Worked explanation

Inventory sp² carbon and sp carbon; can a p orbital used in a hybrid also be counted as a separate residual p?

Follow the operation, its justification and its result.

1. Three sigma directions do not consume the remaining perpendicular p orbital.
2. No double use. Totals remain four in both cases.
3. HCHO carbon has three-domain sp² accounting, three sp² sigma directions and one unhybridised p orbital; C=O has one sigma plus one pi.

Result

No double use. Totals remain four in both cases.

Check

Changed check: for an sp carbon, two hybrids plus two residual p orbitals still total four basis functions.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- A carbon p function can be counted in a hybrid and as a residual orbital.
- Page orientation is the same as relative orbital orientation.
- A localized CO₂ drawing uniquely measures the terminal oxygen basis.
- Three C-N bond lines supply three C-centred directions.
- An input p function can be spent twice in the orbital inventory.

Diagnose

- Inventory sp^2 carbon and sp carbon; can a p orbital used in a hybrid also be counted as a separate residual p ?
- Rotate the whole ethene picture, then rotate only one CH_2 group. Which operation changes relative p alignment?
- Give carbon orbital inventory for CO_2 and HCN . Does a localized drawing uniquely measure the oxygen orbital basis?
- Compare domain counts at C and N with the total number of shared pairs in HCN .
- Build an allocation ledger for sp , sp^2 , sp^3 and flag any duplicate-use entry.

Repair

- $HCHO$ carbon has three-domain sp^2 accounting, three sp^2 sigma directions and one unhybridised p orbital; $C=O$ has one sigma plus one π .
- Ethene has a coplanar sigma framework; the two residual p directions are parallel and perpendicular to it in the introductory model.
- CO_2 carbon: two domains/ sp , two sigma components and two π components. A localized VB drawing is useful accounting, not a unique experimentally compulsory terminal-O orbital-plane picture.
- HCN carbon and nitrogen have introductory sp local accounting; $C-N$ contains one sigma and two π components.
- sp^2 accounting is three hybrid orbitals plus one residual p . A p function already mixed into a hybrid is not simultaneously counted as an additional residual orbital.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: for an sp carbon, two hybrids plus two residual p orbitals still total four basis functions.

2 • APPLY

Inventory HCN carbon without copying the formaldehyde count.

3 • CONNECT

Three sigma directions do not consume the remaining perpendicular p orbital.

2

Construct the relation between the molecular plane and the π orbitals.

Hard concept

Section 5 of 11

← Previous

Next →

Why this section matters for the question set

This unit builds toward

- Q3 · D2

The move learners miss: Align the p functions relative to the molecular plane. [Try it in Core2](#)

1 Build the two local carbon sigma frames.

Why valid: Each ethene carbon has three sigma directions and one remaining p function in sp^2 accounting.

Result: Two in-plane three-direction frames; one perpendicular p at each C.

2 The step the hard question turns on Align the p functions relative to the molecular plane.

Why valid: A residual p is perpendicular to its carbon sigma plane.

Result: Parallel perpendicular p directions in planar ethene.

3 Draw both axial and lateral overlap.

Why valid: The C-C sigma component is axial; lateral p overlap lies above and below the plane.

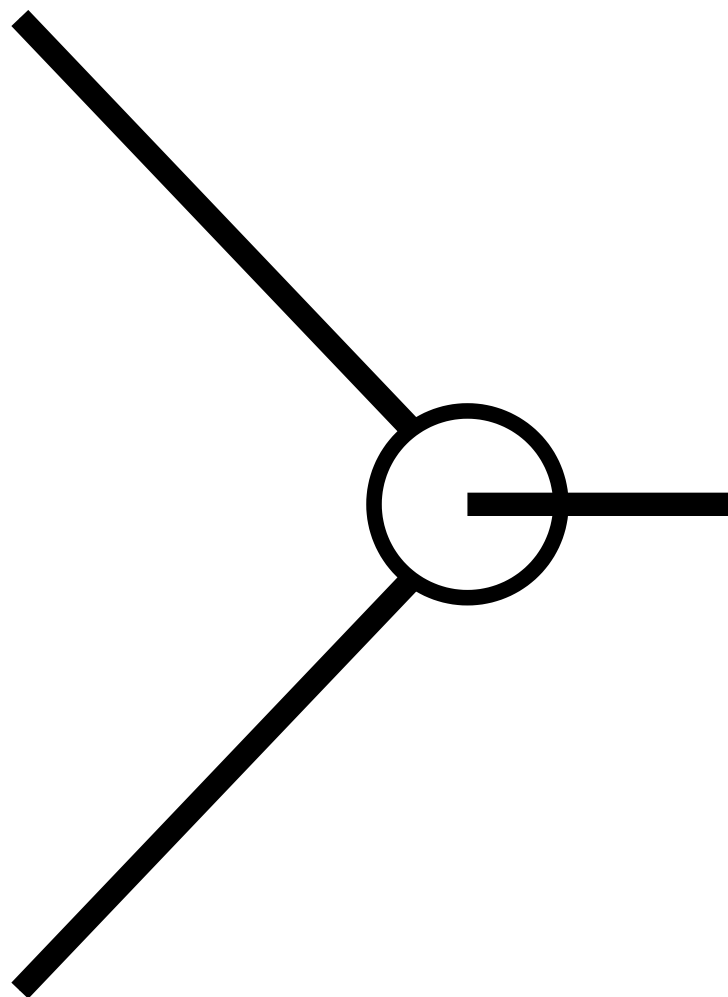
Result: A sigma framework plus a pi component with a node in the molecular plane.

4 Compare rigid rotation with relative CH₂-group rotation.

Why valid: Only the latter changes relative p alignment.

Result: Whole-picture rotation preserves overlap geometry; relative twisting can reduce pi alignment.

Stage 1 — sigma-direction frame



page = molecular plane

Place three directions in one plane.

Stage 1 of 4

Build a generic three-direction sigma frame and add a perpendicular p direction for side-on overlap without reusing orbitals.

Worked explanation

Rotate the whole ethene picture, then rotate only one CH₂ group. Which operation changes relative p alignment?

Follow the operation, its justification and its result.

1. Three sigma directions lie in each sp² carbon plane; residual p is perpendicular to that plane.
2. Only the relative-group rotation changes alignment. A whole-page-orientation answer exposes a representation error.
3. Ethene has a coplanar sigma framework; the two residual p directions are parallel and perpendicular to it in the introductory model.

Result

Only the relative-group rotation changes alignment. A whole-page-orientation answer exposes a representation error.

Check

Changed check: rotate one CH₂ group while preserving C-C connectivity; ask which overlap is sensitive to the relative orientation.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- A carbon p function can be counted in a hybrid and as a residual orbital.
- Page orientation is the same as relative orbital orientation.
- A localized CO₂ drawing uniquely measures the terminal oxygen basis.
- Three C-N bond lines supply three C-centred directions.
- An input p function can be spent twice in the orbital inventory.

Diagnose

- Inventory sp² carbon and sp carbon; can a p orbital used in a hybrid also be counted as a separate residual p?

- Rotate the whole ethene picture, then rotate only one CH₂ group. Which operation changes relative p alignment?
- Give carbon orbital inventory for CO₂ and HCN. Does a localized drawing uniquely measure the oxygen orbital basis?
- Compare domain counts at C and N with the total number of shared pairs in HCN.
- Build an allocation ledger for sp, sp², sp³ and flag any duplicate-use entry.

Repair

- HCHO carbon has three-domain sp² accounting, three sp² sigma directions and one unhybridised p orbital; C=O has one sigma plus one pi.
- Ethene has a coplanar sigma framework; the two residual p directions are parallel and perpendicular to it in the introductory model.
- CO₂ carbon: two domains/sp, two sigma components and two pi components. A localized VB drawing is useful accounting, not a unique experimentally compulsory terminal-O orbital-plane picture.
- HCN carbon and nitrogen have introductory sp local accounting; C-N contains one sigma and two pi components.
- sp² accounting is three hybrid orbitals plus one residual p. A p function already mixed into a hybrid is not simultaneously counted as an additional residual orbital.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: rotate one CH₂ group while preserving C-C connectivity; ask which overlap is sensitive to the relative orientation.

2 • APPLY

Sketch a qualitative twisted ethene comparison without predicting a numerical barrier.

3 • CONNECT

Three sigma directions lie in each sp² carbon plane; residual p is perpendicular to that plane.

Why this section matters for the question set —**This unit builds toward**

- **Q4 · D2**

The move learners miss: Allocate carbon orbitals without duplication.

[Try it in Core2](#)

1 Count two O neighbours at CO₂ carbon.

Why valid: Each C=O multiple bond contributes one local direction.

Result: Two carbon domains; linear introductory framework.

2 The step the hard question turns on Allocate carbon orbitals without duplication.

Why valid: One 2s plus one 2p yields two sp hybrids; two p directions remain perpendicular.

Result: Two sigma directions and two residual C p functions.

3 Account for each localized C=O component.

Why valid: Two double bonds require two sigma and two pi components in this bookkeeping model.

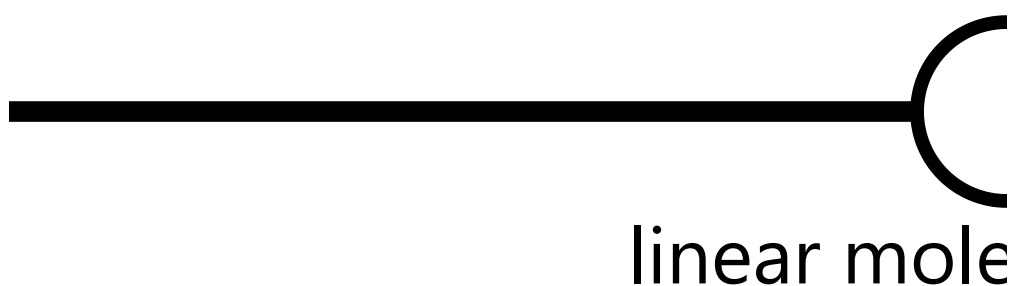
Result: Two sigma + two pi; all four carbon functions have declared roles.

4 Bound the inference about terminal oxygen.

Why valid: A localized basis drawing is one representation of the electronic description; carbon counting does not uniquely measure an oxygen basis.

Result: Linear carbon accounting is consistent; a unique observed O orbital-plane picture is not established.

Stage 1 — two opposite domain / s



Hold two opposite domain directions.

Stage 1 of 4

Coordinate a two-direction linear sigma axis with two independent p directions.

Worked explanation

Give carbon orbital inventory for CO₂ and HCN. Does a localized drawing uniquely measure the oxygen orbital basis?

Follow the operation, its justification and its result.

1. sp accounting leaves two mutually perpendicular p directions.
2. Both carbon inventories close at four. No unique O basis follows from the drawing alone.
3. CO₂ carbon: two domains/sp, two sigma components and two pi components. A localized VB drawing is useful accounting, not a unique

experimentally compulsory terminal-O orbital-plane picture.

Result

Both carbon inventories close at four. No unique O basis follows from the drawing alone.

Check

Changed check: HCN carbon also has two sigma directions and two residual p directions although its neighbours differ.

Mistake clinic • diagnose and repair

—

A tempting wrong path

- A carbon p function can be counted in a hybrid and as a residual orbital.
- Page orientation is the same as relative orbital orientation.
- A localized CO₂ drawing uniquely measures the terminal oxygen basis.
- Three C-N bond lines supply three C-centred directions.
- An input p function can be spent twice in the orbital inventory.

Diagnose

- Inventory sp² carbon and sp carbon; can a p orbital used in a hybrid also be counted as a separate residual p?
- Rotate the whole ethene picture, then rotate only one CH₂ group. Which operation changes relative p alignment?
- Give carbon orbital inventory for CO₂ and HCN. Does a localized drawing uniquely measure the oxygen orbital basis?
- Compare domain counts at C and N with the total number of shared pairs in HCN.
- Build an allocation ledger for sp, sp², sp³ and flag any duplicate-use entry.

Repair

- HCHO carbon has three-domain sp² accounting, three sp² sigma directions and one unhybridised p orbital; C=O has one sigma plus one pi.
- Ethene has a coplanar sigma framework; the two residual p directions are parallel and perpendicular to it in the introductory model.

- CO₂ carbon: two domains/sp, two sigma components and two pi components. A localized VB drawing is useful accounting, not a unique experimentally compulsory terminal-O orbital-plane picture.
- HCN carbon and nitrogen have introductory sp local accounting; C-N contains one sigma and two pi components.
- sp² accounting is three hybrid orbitals plus one residual p. A p function already mixed into a hybrid is not simultaneously counted as an additional residual orbital.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: HCN carbon also has two sigma directions and two residual p directions although its neighbours differ.

2 • APPLY

State what CO₂ geometry supports and what it cannot uniquely prove.

3 • CONNECT

sp accounting leaves two mutually perpendicular p directions.

4

Keep local inventories separate from the triple-bond component inventory.

Hard concept

Section 7 of 11

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q7 • D2

The move learners miss: Allocate each local four-function valence basis.

[Try it in Core2](#)

1 Inventory carbon and nitrogen separately in HCN.

Why valid: C has H and N neighbours; N has one C neighbour plus one lone

pair.

Result: C: two domains; N: two domains.

2 The step the hard question turns on Allocate each local four-function valence basis.

Why valid: Two sp hybrids leave two perpendicular p functions at each second-period centre.

Result: Introductory sp at C and N.

3 Count triple-bond components without adding local directions.

Why valid: Axial overlap gives one sigma; the two residual p pairs give two pi components.

Result: C-N: one sigma + two pi; C-H adds one sigma.

4 Close the ten-valence-electron inventory.

Why valid: H+C+N contributes $1+4+5=10$; four shared pairs and one N lone pair use five pairs.

Result: Eight bonding electrons + two lone-pair electrons = ten.

Stage 1 — two opposite domain / s



Hold two opposite domain directions.

Stage 1 of 4

Coordinate a two-direction linear sigma axis with two independent p directions.

Worked explanation

Compare domain counts at C and N with the total number of shared pairs in HCN.

Follow the operation, its justification and its result.

1. One bonded neighbour supplies one direction even when three bond components are present.
2. Both local domain counts are two; four shared pairs plus N lone pair close the ten-electron ledger.
3. HCN carbon and nitrogen have introductory sp local accounting; C-N contains one sigma and two pi components.

Result

Both local domain counts are two; four shared pairs plus N lone pair close the ten-electron ledger.

Check

Changed check: total valence electrons $\text{HCN} = 1 + 4 + 5 = 10$; the three shared pairs and N lone pair plus C-H pair use all ten.

Mistake clinic • diagnose and repair

A tempting wrong path

- A carbon p function can be counted in a hybrid and as a residual orbital.
- Page orientation is the same as relative orbital orientation.
- A localized CO_2 drawing uniquely measures the terminal oxygen basis.
- Three C-N bond lines supply three C-centred directions.
- An input p function can be spent twice in the orbital inventory.

Diagnose

- Inventory sp^2 carbon and sp carbon; can a p orbital used in a hybrid also be counted as a separate residual p?
- Rotate the whole ethene picture, then rotate only one CH_2 group. Which operation changes relative p alignment?
- Give carbon orbital inventory for CO_2 and HCN . Does a localized drawing uniquely measure the oxygen orbital basis?
- Compare domain counts at C and N with the total number of shared pairs in HCN .
- Build an allocation ledger for sp , sp^2 , sp^3 and flag any duplicate-use entry.

Repair

- HCHO carbon has three-domain sp^2 accounting, three sp^2 sigma directions and one unhybridised p orbital; $\text{C}=\text{O}$ has one sigma plus one pi.
- Ethene has a coplanar sigma framework; the two residual p directions are parallel and perpendicular to it in the introductory model.
- CO_2 carbon: two domains/ sp , two sigma components and two pi components. A localized VB drawing is useful accounting, not a unique

experimentally compulsory terminal-O orbital-plane picture.

- HCN carbon and nitrogen have introductory sp local accounting; C-N contains one sigma and two pi components.
- sp² accounting is three hybrid orbitals plus one residual p. A p function already mixed into a hybrid is not simultaneously counted as an additional residual orbital.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: total valence electrons HCN = 1+4+5 = 10; the three shared pairs and N lone pair plus C-H pair use all ten.

2 • APPLY

Compare HCN and CO₂ local carbon inventories.

3 • CONNECT

One bonded neighbour supplies one direction even when three bond components are present.

5

Avoid spending a carbon basis function twice.

Hard concept

Section 8 of 11

← Previous

Next →

Why this section matters for the question set

—

This unit builds toward

- Q9 · D2

The move learners miss: Mix only the functions assigned to sigma directions. [Try it in Core2](#)

1 Start from four carbon valence-basis functions.

Why valid: The second-period basis is one 2s plus three 2p.

Result: Four inputs; three sigma directions and one pi role required.

2 The step the hard question turns on Mix only the functions assigned to sigma directions.

Why valid: One s plus two p makes three in-plane sp^2 hybrids.

Result: Three hybrids; one p remains outside the mixing.

3 Assign the residual p to the pi role.

Why valid: It is perpendicular to the sigma plane and can overlap laterally with a suitable neighbour.

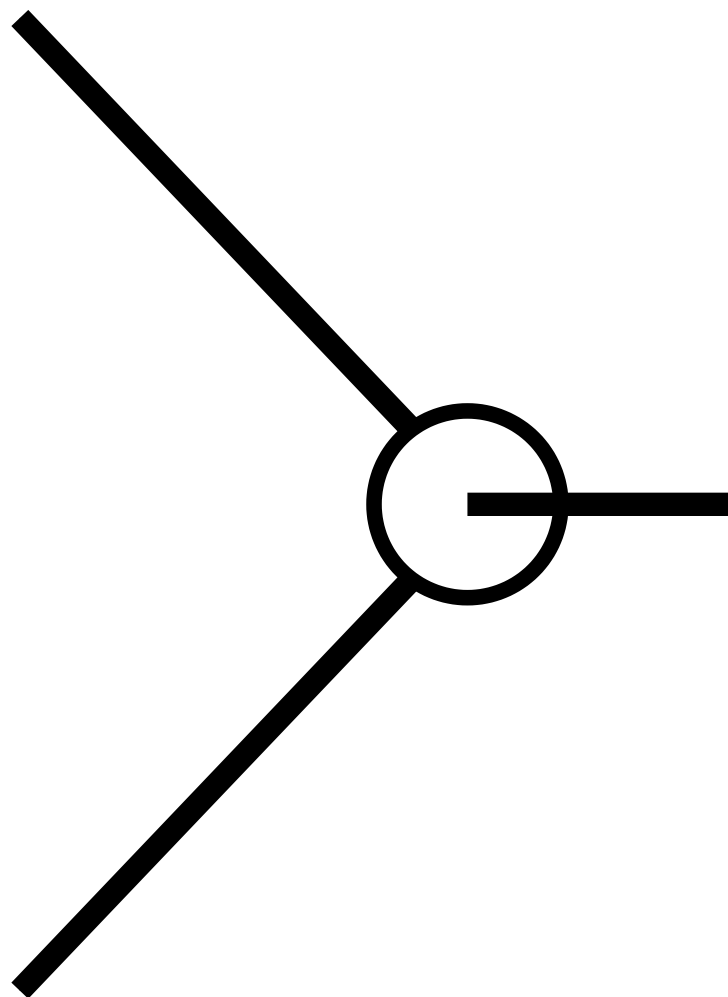
Result: Three sigma hybrids + one residual perpendicular p.

4 Check against sp and sp^3 allocations.

Why valid: Those alternatives account for two hybrids plus two p, or four hybrids plus zero p.

Result: All totals are four; no function is simultaneously a hybrid input and an extra residual p.

Stage 1 — sigma-direction frame



page = molecular plane

Place three directions in one plane.

Stage 1 of 4

Build a generic three-direction sigma frame and add a perpendicular p direction for side-on overlap without reusing orbitals.

Worked explanation

Build an allocation ledger for sp, sp², sp³ and flag any duplicate-use entry.

Follow the operation, its justification and its result.

1. For second-period carbon, one 2s and three 2p functions form a four-dimensional valence basis.
2. All totals four; residual p counts 2,1,0.
3. sp² accounting is three hybrid orbitals plus one residual p. A p function already mixed into a hybrid is not simultaneously counted as an additional residual orbital.

Result

All totals four; residual p counts 2,1,0.

Check

Changed check: sp accounting gives two hybrids plus two residual p; sp³ gives four hybrids plus zero residual p.

Mistake clinic • diagnose and repair

A tempting wrong path

- A carbon p function can be counted in a hybrid and as a residual orbital.
- Page orientation is the same as relative orbital orientation.
- A localized CO₂ drawing uniquely measures the terminal oxygen basis.
- Three C-N bond lines supply three C-centred directions.
- An input p function can be spent twice in the orbital inventory.

Diagnose

- Inventory sp² carbon and sp carbon; can a p orbital used in a hybrid also be counted as a separate residual p?
- Rotate the whole ethene picture, then rotate only one CH₂ group. Which operation changes relative p alignment?

- Give carbon orbital inventory for CO₂ and HCN. Does a localized drawing uniquely measure the oxygen orbital basis?
- Compare domain counts at C and N with the total number of shared pairs in HCN.
- Build an allocation ledger for sp, sp², sp³ and flag any duplicate-use entry.

Repair

- HCHO carbon has three-domain sp² accounting, three sp² sigma directions and one unhybridised p orbital; C=O has one sigma plus one pi.
- Ethene has a coplanar sigma framework; the two residual p directions are parallel and perpendicular to it in the introductory model.
- CO₂ carbon: two domains/sp, two sigma components and two pi components. A localized VB drawing is useful accounting, not a unique experimentally compulsory terminal-O orbital-plane picture.
- HCN carbon and nitrogen have introductory sp local accounting; C-N contains one sigma and two pi components.
- sp² accounting is three hybrid orbitals plus one residual p. A p function already mixed into a hybrid is not simultaneously counted as an additional residual orbital.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: sp accounting gives two hybrids plus two residual p; sp³ gives four hybrids plus zero residual p.

2 • APPLY

Allocate the sp² basis without memorising an orbital picture.

3 • CONNECT

For second-period carbon, one 2s and three 2p functions form a four-dimensional valence basis.

Question-specific diagnosis and repair

—

Repair for Q2

Test the rule on a changed case

Inventory sp^2 carbon and sp carbon; can a p orbital used in a hybrid also be counted as a separate residual p?

Your prediction and reason

Repair for Q3

Test the rule on a changed case

Rotate the whole ethene picture, then rotate only one CH_2 group. Which operation changes relative p alignment?

Your prediction and reason

Repair for Q4

Test the rule on a changed case

Give carbon orbital inventory for CO_2 and HCN . Does a localized drawing uniquely measure the oxygen orbital basis?

Your prediction and reason

Repair for Q7

Test the rule on a changed case

Compare domain counts at C and N with the total number of shared pairs in HCN.

Your prediction and reason

Repair for Q9

Test the rule on a changed case

Build an allocation ledger for sp, sp², sp³ and flag any duplicate-use entry.

Your prediction and reason

COMPETITION TRANSFER

Close an orbital inventory under a changed local description

Original transfer task; not claimed as an official exam/PYQ item.

A neutral carbon centre is described by three coplanar sigma-bond directions and one perpendicular p-orbital participation. Without using a memorised molecule name, give an orbital-job inventory and state one check that would expose double-counting.

Check your reasoning

Three sigma-framework hybrid directions and one separate p-orbital pi job; check that each local valence orbital is allocated once.

1. The three sigma directions define the local sigma framework.
2. The perpendicular p direction is reserved for side-on pi overlap.
3. A closure check prevents any orbital from being assigned twice.

Now you do one

For $\text{H-C}\equiv\text{C-H}$, account for each carbon valence basis and the whole-molecule sigma/pi inventory without reusing a p function.

Model answer

Practice this concept in Core2

Q2

Q3

Q4

Q7

Q9

Sections

Compare full local inventories, not only the common three sigma bonds.

Count local domains and bonded nuclei at two different centres.

Test whether domain geometry and nuclear shape are interchangeable.

Electron-domain geometry versus molecular shape

Subject: Chemistry

Topic: Chemical bonding and hybridisation

Concept: Electron-domain geometry versus molecular shape

Concept difficulty: Medium concept

1

Compare full local inventories, not only the common three sigma bonds.

Medium concept

Conditions for this construction

- Can count central domains.
- Knows the four-domain tetrahedral rule.

Why this section matters for the question set

This unit builds toward

- **Q1** · D2

The move learners miss: Apply the bounded introductory domain models.

[Try it in Core2](#)

1 Inventory bonds and central lone pairs separately.

Why valid: The givens supply three bonds at each centre but different lone-pair counts.

Result: BF₃: 3+0=3 domains; NH₃: 3+1=4.

2 The step the hard question turns on Apply the bounded introductory domain models.

Why valid: The local total, rather than bonds alone, selects the model arrangement.

Result: BF₃: planar/sp²; NH₃: tetrahedral domains/sp³-like.

3 Name shape using bonded nuclei only.

Why valid: The N lone-pair domain has no bonded nucleus.

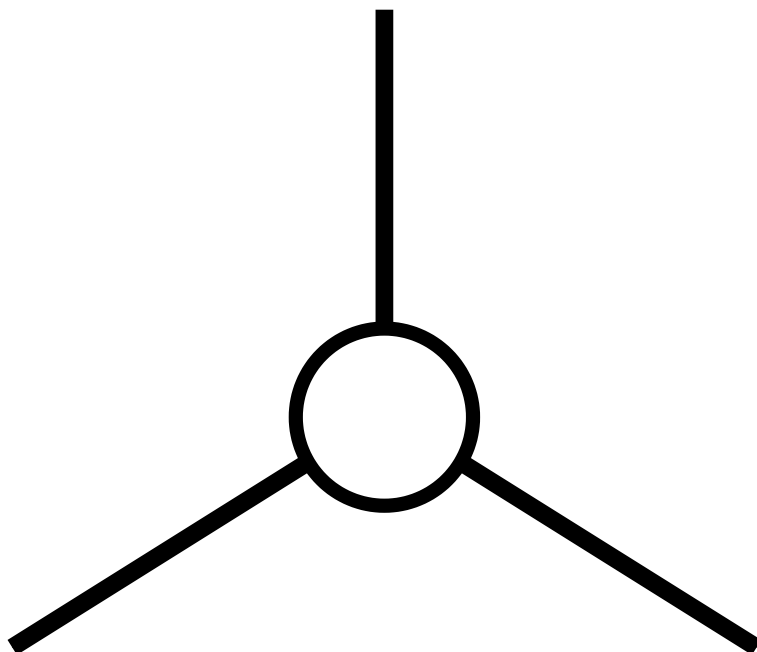
Result: BF₃ is trigonal planar; NH₃ is trigonal pyramidal.

4 Compare NH₃ with H₂O.

Why valid: Three-bond/one-pair and two-bond/two-pair inventories both total four.

Result: The same domain total can give different bonded-nucleus shapes.

Count directions / regions first



3 domain directions

Count regions before labels.

Stage 1 of 3

Separate electron-domain directions from atom-only molecular shape and make lone-pair occupancy visible.

The key step

CH₄, NH₃, H₂O have four introductory domains but tetrahedral, pyramidal and bent nuclear shapes respectively.

Worked explanation

Count local domains and bonded nuclei separately in BF₃, NH₃ and H₂O.

Follow the operation, its justification and its result.

1. A central lone pair adds a domain even though it adds no bonded nucleus.
2. Totals 3,4,4; bonded nuclei 3,3,2.
3. BF₃ has three B domains; NH₃ has three bonds plus one N lone pair. Introductory sp²/planar versus sp³-like domain/pyramidal shape follows the local model.

Result

Totals 3,4,4; bonded nuclei 3,3,2.

Check

Changed check: H₂O and NH₃ have different numbers of bonds but the same four-domain total.

Mistake clinic · diagnose and repair

A tempting wrong path

- Three sigma bonds always imply three domains.
- One whole-molecule label determines both carbon and nitrogen shapes.
- Four-domain geometry equals the atom-only shape in every case.

Diagnose

- Count local domains and bonded nuclei separately in BF₃, NH₃ and H₂O.
- Compare local N inventory with carbon inventory in CH₃NH₂. Which centre includes a lone-pair direction?
- For four-domain inventories with 4,3,2 bonded atoms, give the atom-only shape and explain omitted slots.

Repair

- BF₃ has three B domains; NH₃ has three bonds plus one N lone pair. Introductory sp²/planar versus sp³-like domain/pyramidal shape follows the local model.
- Methylamine C has four bonded directions; N has three bonded directions and one lone pair. Introductory tetrahedral C and pyramidal N nuclear arrangements differ despite four-domain totals.

- CH₄, NH₃, H₂O have four introductory domains but tetrahedral, pyramidal and bent nuclear shapes respectively.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: H₂O and NH₃ have different numbers of bonds but the same four-domain total.

2 • APPLY

Compare a four-bond centre with a two-bond/two-lone-pair centre.

3 • CONNECT

A central lone pair adds a domain even though it adds no bonded nucleus.

2

Count local domains and bonded nuclei at two different centres.

Medium concept

Section 10 of 11

← Previous

Next →

Why this section matters for the question set

This unit builds toward

- Q8 · D2

The move learners miss: Assign the localized introductory domain models.

[Try it in Core2](#)

1 Make atom-local inventories in CH₃NH₂.

Why valid: C has four single bonds; N has three single bonds and one lone pair.

Result: C: 4+0; N: 3+1; both total four domains.

2 The step the hard question turns on Assign the localized introductory domain models.

Why valid: Each four-domain centre can use sp³-like accounting under the

stated local model.

Result: Four model directions at each centre.

3 Count nuclei for the separate shape descriptions.

Why valid: C has four bonded neighbours while N has three; a lone pair is not an atomic vertex.

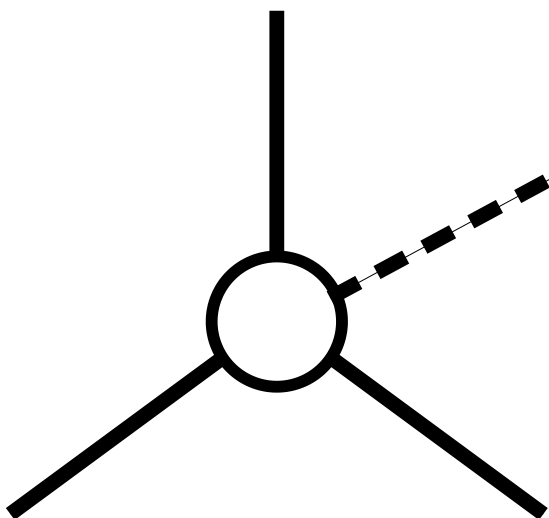
Result: Tetrahedral C nuclear arrangement; pyramidal N arrangement.

4 Replace an N-H bond with an N-CH₃ bond in a neutral amine comparison.

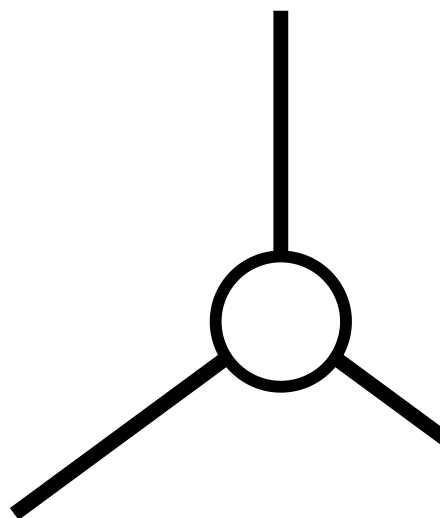
Why valid: The N inventory stays three bonds plus one lone pair.

Result: Local N count and pyramidal introductory description persist; exact angle is unresolved.

Same four-domain scaffold



4 bond positions



3 bonds + 1 lo

Compare 4 bonds, 3+1 and 2+2.

Stage 1 of 3

Project the same four-domain scaffold to different bonded-atom shapes as lone-pair occupancy changes.

Worked explanation

Compare local N inventory with carbon inventory in CH_3NH_2 . Which centre includes a lone-pair direction?

Follow the operation, its justification and its result.

1. A lone pair contributes to electron-domain geometry but is excluded from molecular-shape naming.
2. N does; C does not. Using one atom-global shape for both reveals locality confusion.
3. Methylamine C has four bonded directions; N has three bonded directions and one lone pair. Introductory tetrahedral C and pyramidal N nuclear arrangements differ despite four-domain totals.

Result

N does; C does not. Using one atom-global shape for both reveals locality confusion.

Check

Changed check: replacing N-H with N-CH₃ preserves the local three-bond-plus-lone-pair N inventory in the neutral localized amine model.

Mistake clinic · diagnose and repair

A tempting wrong path

- Three sigma bonds always imply three domains.
- One whole-molecule label determines both carbon and nitrogen shapes.
- Four-domain geometry equals the atom-only shape in every case.

Diagnose

- Count local domains and bonded nuclei separately in BF_3 , NH_3 and H_2O .

- Compare local N inventory with carbon inventory in CH_3NH_2 . Which centre includes a lone-pair direction?
- For four-domain inventories with 4,3,2 bonded atoms, give the atom-only shape and explain omitted slots.

Repair

- BF_3 has three B domains; NH_3 has three bonds plus one N lone pair. Introductory sp^2 /planar versus sp^3 -like domain/pyramidal shape follows the local model.
- Methylamine C has four bonded directions; N has three bonded directions and one lone pair. Introductory tetrahedral C and pyramidal N nuclear arrangements differ despite four-domain totals.
- CH_4 , NH_3 , H_2O have four introductory domains but tetrahedral, pyramidal and bent nuclear shapes respectively.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: replacing N-H with N- CH_3 preserves the local three-bond-plus-lone-pair N inventory in the neutral localized amine model.

2 • APPLY

Describe a neutral secondary amine N without predicting an exact experimental angle.

3 • CONNECT

A lone pair contributes to electron-domain geometry but is excluded from molecular-shape naming.

3

Test whether domain geometry and nuclear shape are interchangeable.

Medium concept

Section 11 of 11

← Previous

Why this section matters for the question set

—

This unit builds toward

- **Q10** · D2

The move learners miss: Use all domains for domain geometry.

[Try it in Core2](#)

1 Hold the four-domain total fixed across three cases.

Why valid: CH₄, NH₃ and H₂O have respective bond/pair inventories 4/0, 3/1 and 2/2.

Result: All have four introductory domains.

2 **The step the hard question turns on** Use all domains for domain geometry.

Why valid: Bonding and lone-pair entries occupy the domain-model directions.

Result: Tetrahedral electron-domain arrangement in all three.

3 Use bonded nuclei for molecular shape.

Why valid: Removing LP entries from the shape name changes the number of atomic vertices.

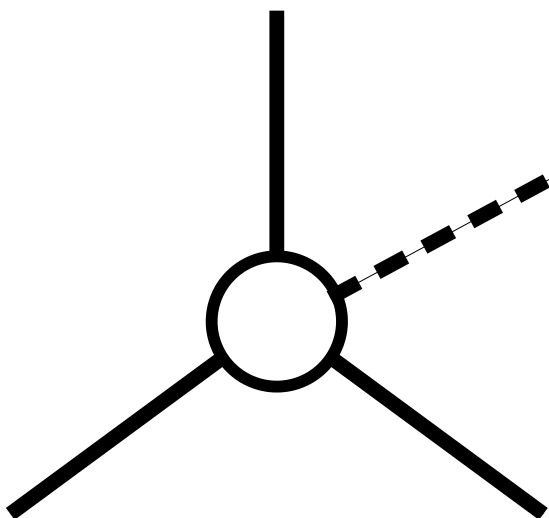
Result: CH₄ tetrahedral; NH₃ pyramidal; H₂O bent.

4 State the condition for the two names to coincide.

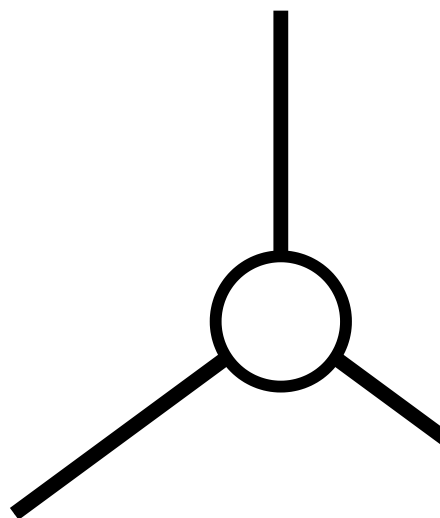
Why valid: When there is no central lone pair every domain direction terminates at a bonded nucleus.

Result: The student claim holds for the no-lone-pair case, not universally.

Same four-domain scaffold



4 bond positions



3 bonds + 1 lo

Compare 4 bonds, 3+1 and 2+2.

Stage 1 of 3

Project the same four-domain scaffold to different bonded-atom shapes as lone-pair occupancy changes.

Worked explanation

For four-domain inventories with 4,3,2 bonded atoms, give the atom-only shape and explain omitted slots.

Follow the operation, its justification and its result.

1. A lone-pair domain has no bonded nucleus to include in molecular-shape naming.
2. Tetrahedral, pyramidal, bent; omitted slots are lone pairs. Repeating tetrahedral for all atom-only shapes exposes the conflation.

3. CH_4 , NH_3 , H_2O have four introductory domains but tetrahedral, pyramidal and bent nuclear shapes respectively.

Result

Tetrahedral, pyramidal, bent; omitted slots are lone pairs. Repeating tetrahedral for all atom-only shapes exposes the conflation.

Check

Changed check: a four-bond/no-lone-pair centre cannot be assigned the same atom-only shape as a two-bond/two-lone-pair centre.

Mistake clinic · diagnose and repair

A tempting wrong path

- Three sigma bonds always imply three domains.
- One whole-molecule label determines both carbon and nitrogen shapes.
- Four-domain geometry equals the atom-only shape in every case.

Diagnose

- Count local domains and bonded nuclei separately in BF_3 , NH_3 and H_2O .
- Compare local N inventory with carbon inventory in CH_3NH_2 . Which centre includes a lone-pair direction?
- For four-domain inventories with 4,3,2 bonded atoms, give the atom-only shape and explain omitted slots.

Repair

- BF_3 has three B domains; NH_3 has three bonds plus one N lone pair. Introductory sp^2 /planar versus sp^3 -like domain/pyramidal shape follows the local model.
- Methylamine C has four bonded directions; N has three bonded directions and one lone pair. Introductory tetrahedral C and pyramidal N nuclear arrangements differ despite four-domain totals.
- CH_4 , NH_3 , H_2O have four introductory domains but tetrahedral, pyramidal and bent nuclear shapes respectively.

1-2-3 QUICK CHECK

1 • CHECK

Changed check: a four-bond/no-lone-pair centre cannot be assigned the same atom-only shape as a two-bond/two-lone-pair centre.

2 • APPLY

Sketch the domain and atom-only inventories separately for H₂O.

3 • CONNECT

A lone-pair domain has no bonded nucleus to include in molecular-shape naming.

Question-specific diagnosis and repair

Repair for Q1

Test the rule on a changed case

Count local domains and bonded nuclei separately in BF₃, NH₃ and H₂O.

Your prediction and reason

Repair for Q8

Test the rule on a changed case

Compare local N inventory with carbon inventory in CH₃NH₂. Which centre includes a lone-pair direction?

Your prediction and reason

Repair for Q10

Test the rule on a changed case

For four-domain inventories with 4,3,2 bonded atoms, give the atom-only shape and explain omitted slots.

Your prediction and reason

COMPETITION TRANSFER

Hold domain geometry fixed while changing atom occupancy

Original transfer task; not claimed as an official exam/PYQ item.

A four-domain centre is changed from four bonds to two bonds plus two lone pairs without changing the total domain count. Explain which geometry description is preserved and which atom-only description must be reconsidered.

Check your reasoning

The four-domain electron geometry stays tetrahedral; the molecular shape must be derived from the two bonded-atom positions.

1. All four electron domains remain.
2. Molecular shape omits lone-pair positions.

Now you do one

Given OF₂ with two O-F bonds and two O lone pairs, sketch the four domain directions and distinguish its atom-only shape from the domain arrangement.

Model answer

Practice this concept in Core2

Q1

Q8

Q10

Chemistry · Chemical bonding & hybridisation — Issue #33 candidate